Lab 2 - 4: The Public Showcase (Static Website Hosting)

1. Introduction & Scenario

The "Project Starlight" team has captured a breathtaking "Image of the Month" of the Eagle Nebula. It's a stunning piece of work, and the university wants to share it with the world. A traditional web server would be too complex and costly to set up for just one page.

Your final task as the project's data manager is to leverage S3 to host a simple, beautiful, and serverless public webpage to showcase this achievement. This lab will walk you through configuring an S3 bucket to serve web content directly to the internet, one of the most popular and powerful features of S3.

2. Learning Objectives

By the end of this lab, you will be able to:

- Configure an S3 bucket for static website hosting.
- Author and apply a public-read bucket policy in JSON.
- Upload a directory of website files to S3 using the `sync` command.
- Access and test a live, publicly accessible website hosted on S3.

3. Prerequisites

- An AWS account and access to AWS CloudShell.

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4. Lab Steps

Step 1: Prepare the Website Files in CloudShell

First, we need to create the HTML, CSS, and image files for our showcase. We will do this entirely within CloudShell.

1. Launch **AWS CloudShell**.
2. Create a new directory for the website and move into it.

   ```
   mkdir public-showcase && cd public-showcase
   ```

3. Create the main HTML file using the `nano` text editor.

   ```
   nano index.html
   ```

4. Copy and paste the following HTML code into the `nano` editor:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Project Starlight Showcase</title>
```

```
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <h1>Image of the Month</h1>
        <h2>The Eagle Nebula</h2>
        <img src="eagle-nebula.jpg" alt="The Eagle Nebula">
        <p>This stunning image was captured by the Project Starlight team on September 17, 2025.</p>
    </div>
</body>
</html>
```

5. Save and exit `nano` by pressing `Ctrl+X`, then `Y`, then `Enter`.
6. Now, create the stylesheet.

   ```
   nano style.css
   ```

7. Copy and paste the following CSS code into the editor:

```
body {
    font-family: Arial, sans-serif;
    background-color: #111;
    color: #eee;
    margin: 0;
    padding: 20px;
}
.container {
    max-width: 800px;
    margin: 0 auto;
    text-align: center;
}
img {
    max-width: 100%;
    border-radius: 8px;
    margin-top: 20px;
```

```
}
```

8. Save and exit `nano`.
9. Finally, download the beautiful nebula image.

```
wget https://esahubble.org/media/archives/images/large/heic1501a.jpg -O eagle-nebula.jpg
```

You now have a complete, self-contained website in your `public-showcase` directory.

Step 2: Create and Configure the Bucket for Website Hosting

1. Create a new, globally unique bucket. **Remember to update `yourname` and the date.**

```
BUCKET_NAME="starlight-showcase-yourname-20250917"

aws s3 mb s3://$BUCKET_NAME
```

2. Use the `aws s3 website` command to configure the bucket. The `--index-document` flag tells S3 which file to serve as the homepage.

```
aws s3 website s3://$BUCKET_NAME --index-document index.html
```

Step 3: Create the Public Access Bucket Policy

By default, all S3 buckets are private. To make a website public, you must attach a bucket policy that explicitly allows `GetObject` access to everyone.

1. Create a new JSON file for the policy.

```
nano bucket-policy.json
```

2. Copy and paste the following policy. This policy says, "Allow anyone (`"Principal": "*"`) to perform the `s3:GetObject` action on any object (`/*`) within this specific bucket."

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR_BUCKET_NAME/*"
    }
  ]
}
```

3. **CRITICAL:** Inside `nano`, replace the placeholder text `YOUR_BUCKET_NAME` with your actual bucket name variable: `$BUCKET_NAME`. The final resource line should look like `"Resource": "arn:aws:s3:::$BUCKET_NAME/*"`.

4. Save and exit `nano`.

Step 4: Apply the Bucket Policy

1. Use the `aws s3api put-bucket-policy` command to attach the policy to your bucket.

   ```
   aws s3api put-bucket-policy --bucket $BUCKET_NAME --policy file://bucket-policy.json
   ```

2. If successful, there is no output. Your bucket is now public.

Step 5: Upload the Website Files

1. Navigate back to the parent directory of your website files.

   ```
   cd ..
   ```

2. Use the `aws s3 sync` command to upload your entire `public-showcase` directory to the S3 bucket. `sync` is perfect for websites because it only copies new or updated files.

   ```
   aws s3 sync public-showcase/ s3://$BUCKET_NAME
   ```

3. You will see output confirming the upload of `index.html`, `style.css`, and `eagle-nebula.jpg`.

Step 6: Find and Test Your Live Website!

1. The URL for an S3 static website follows a standard format. You need to know which region your CloudShell is running in. Find it with this command:

   ```
   aws configure get region
   ```

2. Your website URL is: http://<bucket-name>.s3-website.<region>.amazonaws.com.
   For example, if your region is us-east-1, you can see your URL with:

   ```
   echo "http://$BUCKET_NAME.s3-website.us-east-1.amazonaws.com"
   ```

3. Copy the URL, paste it into your web browser, and press Enter.

You should now see your beautiful, live "Image of the Month" showcase page!

Deliverable

To receive completion marks for this lab, the deliverable is a screenshot of the final, working result. Your screenshot must be of your entire web browser window.

6. Conclusion & Cleanup

Congratulations on completing the Project Starlight S3 module! You have progressed from simple file uploads in the console to deploying a secure, cost-effective, and globally accessible static website entirely from the command line.

1. To clean up, run this single command to delete the bucket and all the files within it:

```
aws s3 rb s3://$BUCKET_NAME --force
```